MIL-G-19053C(AER)
14 January 1957
Superseding
MIL-G-19053B(AER)
23 July 1956

MILITARY SPECIFICATION

GAGE, AIRCRAFT, CAPACITANCE, LIQUID OXYGEN CONVERTER, GENERAL SPECIFICATION FOR

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers the general requirements for a capacitance type liquid oxygen quantity gage system for indicating the quantity of liquid oxygen in an aircraft liquid oxygen system.

1.2 Classification -This specification covers capacitance type gages used to indicate the amount of liquid oxygen contained in aircraft liquid oxygen converters.

2. APPLICABLE COMMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

BB-O-925	Oxygen, 99.5 Percent, Gas and Liquid

Military

MIL-A-8625	Anodic Coatings, for Aluminum Aluminum Alloys
MIL-C-5543	Chemical Films for Aluminum and Aluminum Alloys
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for
MIL-D-5028(ASG)	Drawings and Data Lists; Preparation of Manufacturer's (For Production Aircraft, Guided Missiles, Engines, Accessories and Other Auxiliary Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-F-5400	Electronic Equipment; Airborne, General Specification for
MIL-E-7894(ASG)	Electric Power, Aircraft Characteristics of
MIL-I-6181	Interference Limits and Tests; Aircraft, Electrical and Electronic Equipment
MIL-I-19326(Aer)	Installation of Liquid Oxygen Equipment in Aircraft, General Specification for
MIL-P-5633	Packaging and Packing of Aircraft Material in Steel Shipping Containers
MIL-P-4064	Packaging of Light-weight Aircraft Accessories
MIL-P-6906	Plates, Information and Identification
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-W-16878(Navy)	Wire, Electrical (Insulated, High Temperature)
MIL-L-2567(ASG)	Lighting, Integral, Instrument, General Specification for

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STANDARDS

MIL-STD-129	Marking of Shipments
MIL-STD-130	Identification Marking of U.S. Military Property

DRAWINGS

Air Force - Navy Aeronautical

AN-3140	T-1 3/4 Bulb, Midget Flange Base
AND-10398	Metals -Definition of Dissimilar
AND-10404	Pointers-Standard Design of Aircraft Instrument
AND-10412	Case -2 Inch Size Instrument, Standard Dimensions for

Military

MS-33558	Numerals and Letters, Aircraft Instrumental Dial

PUBLICATIONS

Air Force - Navy Aeronautical Bulletin

No. 143	Specification and Standards, Use of

(When requesting Specifications, standards, drawings, and publications refer to both title and number. Copies of this specification and applicable specifications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT)

3* REQUREMENTS

3.1 Qualification - Gages furnished under this specification shall be products which have been tested, and have passed the qualification tests specified herein.

3.2 Materials - Materials used in the construction of the gages shall conform to applicable specifications and as specified herein. Materials not so covered or so described shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.2.1 Protective Treatment -Materials used in the construction of the gage that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage shall be protected against such corrosion in a manner that will in no way prevent complaince with the performance requirements of this specification.

3.2.2 Metals - Metals shall be of the corrosion resisting type.

3.2.2.1 Dissimilar Metals - Unless suitably protected against electrolytic corrosion dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined by Drawing AND-10398.

3.2.3 Selection of Materials - Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment which are not specifically designated herein and which are necessary for the execution of this specification, shall be selected in accordance with ANA Bulletin No. 143, except as provided in the following paragraph

3.2.3.1 Standard Parts - Standard parts (MS, AN or JAN) shall be used whenever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, etc., may be used provided they possess suitable properties and are replaceable by the standard parts (MS, AN or JAN) without alteration, and provided the corresponding standard part numbers are referenced in the parts list and, if practicable, on the contractor's drawings. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used provided they conform to all requirements of this specification.

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3.3 Design - The gauging system shall be used to indicate the quantity of liquid oxygen contained in the aircraft oxygen system as installed in accordance with MIL-I-19326(Aer). The liquid oxygen used the the aircraft oxygen system shall coform to Federal Specification BB-O-925, Type II. The gaging system shall be a null-balancing, three wire, capacitance type bridge. The indicator shall have provisions for driving a repeater indicator regardless of whether or not there is a requirement for a repeater in aircraft using the liquid oxygen indicator. The indicator shall control the outlet voltage to the repeater indicator which shall not exceed a maximum of 20 volts. The probe of the liquid oxygen oonverter shall. be the measuring capacitor of the capacitance bridge system. The gaging system shall consist of a probe and indicator. By electronic circuits required for amplification and power supply shall be a part of the indicator. The gage shall be adaptable to installation where totalized indication of the quantity of liquid oxygen in two or more converters is required, The gaging system shall be designed in all cases to operate without field adjustments. However, factory adjustment provisions shall be incorporated, Vacuum tubes shall not be used in the electronic circuit of the gage. Semi-conductor devices used shall be capable of satisfactory operation up to 165°F. Ectrudsd Teflon covered wiring, for both shielded and unshielded leads, meeting MIL-W-16878, Type E, AWG size 20. (see Figure 2), shall be used between the probe and indicator, Connectors shall be as shown in Figure 2. Any approved connector jacks which mount on the converter panel to mate with the specified connector plugs on the shielded and unshielded wires may be used. Variation in the length of the lead wire between the probe and indicator shell not affect gage system accuracy. The over-all system accuracy shall be ±2 percent of indication plus ±4 percent of full scale indication at any of the major dial divisions.

3.3.1 Indicator - The indicator shall be hermetically sealed and shall be of the clamp-on type in accordance with Drawing AND-10412 as shown in Figure 1. The indicator shall indicata the actual amount of liquid oxygen contained in the aircraft liquid oxygen converter measured by weight with tolerances as specified in detail specifications. The indicator quantity gradations shall be as shown in detail specifications. Figure 2 shows harness and connectors.

3.3.1.1 Low Liquid Level Warning Circuit - The Indicator shall provide for a low liquid level warning circuit to operate a remote warning system utilizing two AN 3140-327 lamps or two AN 3121-313 lamps at each station where this information is required in the aircraft. The low liquid level limit shall be be prescribed in the applicable detail specification.

3.3.1.2 Press to Test - The indicator shall provide for a press to test circuit whioh will drive the pointer bac to the zero position or below.

3.3.1.3 Power Failure - The indicator shall provide indication of power failure or interruption a manual reset mechanism shall not be incorporated. The presentation in the indicator shall be as shown in detail specifications.

3.3.2 Electrical Parts - All standard parts, suoh as resistors, condensers, and transformers, shall confrom to Specification MIL-E-5400 or equivalent.

3.3.3 Construction - The indicator shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and othsr conditions incident to shipping, storage, installation, and service.

3.3.4 Cases - The indicator case shall be in conformance with Drawing AND-10412 and the applicable detail specification. The case shall be made of drawn metal. suitable for hermetic sealing and of light construction.

3.3.4.1 Case Connector - The case connector shall be in accordance with Figure 2.

3.3.4.2 Filling Medium - The cases shall be fills d with an inert gas composed of at least 10 percent helium and 90 percent nitrogen to provide means for leak detection. The gases shall be of at least 98 percent purity, free of dust particles, and containing not more than 0.006 milligrams of water vapor per liter (dewpoint-65 degrees C) at the filling pressure. The absolute pressure of the gas in the case shall be approximatel y one atmosphere.

3.3.5 Dial - The indicator dial shall be fastened securely to the case or the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The pointer movemmt shall not be limited by a mechanical stope on the dial.

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3.3.5.1 Markings - Dial markings shall conform to the requirements of the detail Specification. The style and proportions of the numerals and letters placed on the dial shall conform to Drawing MS-33558. Numerals shall distinctly indicate the graduation to which they apply.

3.3.6 Cover Glass - The cover glass of the indicator shall be suitable for hermetic sealing and shall be flat, clear, and free from flaws which interfere with the normal reading of the indicator.

3.3.7 Pointer - The pointer length shall be such that the pointer tip will extend to but not overlap the liter graduations, The pointer shall conform to the requirements of Drawing AND-10404-8.

3.3.8 Lighting - An integral lighting system in accordance with MIL-L-25467 shall be provided with the following exceptions:

Light Sources - Lamps conforming to Drawing AN 3140-327 shall be used as the lighting source.

b. Electrical Connectors -Electrical connectors shall be as shown in Figure 2.

3.3.9 Fogging - The gage assembly shall be kept at a temperature of -65°F ± 2°, for at least 4 hours, There shall be no evidence of fogging of the cover glass during this period.

3.3.10 System Power Requirements - The gage system shall be designed to operate from a 115 volt, single phase, 400 cycle A-C power supply. The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein, The power required shall not exceed the specified amounts.

a. AC Power (Single Phase)

(1) 115 Volt Constant Frequency

Operating Voltage Limits	102 to 124 volts
Operating Frequency Limits	380 to 420 cps

3.3.10.1 Overload Protection - Overload protection for the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other devises shall be held to minimum.

3.3.10.2 Under Voltage Protection - The equipment shall not be damaged by voltages below the mimum specified and shall automatically resume normal operation when the voltage return within limits.

3.4 Interchangeability – All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respeot to installation and performance, Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.5 Finishes and Protective Coatings

3.5.1 Finish - Finishes which will crack, chip, or scale during the normal service life or due to extreme of atmospheric conditions, shall not be used.

3.5.1.1 Aluminum Alloy Parts -Aluminum alloy parts, where practicable shall be covered with a protective film conforming to Specification MIL-A-8625 or MIL-C-5541.

3.5.1.2 Steel Parts - All steel parts (except stainless steel) exposed to the atmosphere, shall be corrosion resistant.

3.6 Screw Threads - Unless otherwise specified, the threads of all machine scres 0.060 inch or larger in diameter shall conform to Specification MIL-S-7742.

3.7 Electrical Receptacles (Male and Female) and Wiring - Gage system components shall be provided with electrical. receptacles and wiring as shown in Figure 2.

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3.8 Weight - The weight of the indicator shall not exceed the weight specified in the applicable detail specification.

3.9 Performance - The gage shall be capable of meeting the requirements specified in section 4.

3.9.1 Pointer Overshoot - The gage shall be designed to provide steady indication of the pointer. In approaching the null, the pointer shall not overshoot the final position more than one the and shall return to the final position within 2 seconds.

3.9.2 Response Time - The gage system shall be designed so that tine indicator pointer will traverse full scale from zero in not more than 15 seconds to provide as a means, if required, of indicating when the liquid oxygen system is filled.

3.10 Identification of Product

3.10.1 Nameplate - Nameplates conforming to Specification MIL-P-6906 shall be securely attached to each separately mounted component and shall be legibly and permanently marked with information in accordance with Specification MIL-STD-130 and the applicable detail specification.

3.11 Workmanship - Each item including all parts and accessories, shall be constructed in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, welding, and brazing, painting, riveting, machine screw assemblies, and freedom from burrs and sharp edges,

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests - Inspection and testing shall be classified as follows:

a. Qualification Tests: Qualification tests are those performed on samples submitted for approval as qualified products.

b, Inspection Tests: Inspection tests are those tests performed on items manufactured and submitted for acceptance under contract,

4.2 Qualification Tests.

4.2.1 Instructions- Qualification test samples shall consist of one each item upon which qualification is desired. The samples shall be identified as required and forwarded to the activity responsible for qualification, designated in the letter of authorization from that activity. (See Section 6)

4.2.2 Tests - Qualification tests shall consist of all tests of this specification except where otherwise specified.

4.3 Inspection Tests - Inspection tests shall consist of Individual tests and Sampling tests.

4.3.1 Individual Tests - Individual tests shall be conducted on each item and shall consist of the following tests of this specification and in addition any other tests specified herein which the inspector considers necessary to determine compliance with the requirements of this specification:

a. Examination of Product
b. Indicator Scale Error
c. Sealing
d. Dielectric Test
e. Lighting
f. Low Level Warning
g. Press to Test
h. Power Failure
i. Fogging

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4.3.2 Sampling Tests -Sampling tests shall be conducted on one item selected at random from each lot of 150 or fraction thereof which have passed all Individual tests. The lot shall consist of identical items of the same manufacturer's part number, manufactured under essentially the same conditions. Sampling tests shall consist of the following tests:

a. Temperature Tests of Paragraphs 4.5.$.1 and 4.5.8.2
b. Vibration Tests

4.4 Test Conditions

4.4.1 Standard Atmospheric Conditions for Indicator - Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches of Hg.) room temperature (approximately 25 degrees C), and normal relative humidity (approximately 30 percent). When tests are made with atmospheric pressure, room temperatures, or relative humidity differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.4.2 Tapping - Unless otherwise specified, the components may be lightly tapped before a test reading is taken.

4.4.3 Position -Unless otherwise specified, the components shall be tested in the normal operating position.

4.4.4 Test Power - Unless otherwise specified herein, the components shall be tested with 115 volts, 400 cycle, single-phase, A.C. power supply. The voltage and frequency tolerance shall be plus or minus 2 percent.

4.4.5 Master Test Capacitor -Whenever a master test capacitor is specified, it shall be a precision variable capacitor which shall have an accuracy of 0.2% or 0.2. uuf, whichever is the greater, at a temperature of 25°C. In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement.

4.4.6 Nominal Liquid Oxygen - Nominal liquid oxygen, as specified herein, shall conform to Federal Specification BB-O-925, Type II. The dielectric constant for liquid oxygen at an equilibrium pressure of zero (0) psig shall be taken as 1,483.

4.4.7 Weighing Equipment - The accuracy of the weighing equipment to be used during the cxomplete Gage Scale Error Test shall be ±0.02 pounds. In all subsequent tests and measurements specified herein, the tolerance of the measuring equipment shall be added to the specified tolerance for the given test or measurement.

4.5 Test Methods

4.5.1 Examination of Product - Each item shall be examined to determine conformance to this specification and the applicable detail specification with respect to materials, workmanship, weight, and marking.

4.5.2 Complete Gage Scale Error - The complete gage system shall be electrically connected, with the probe installed in the converter for which it is intended. The test shall be conducted in the following manner:

a. Fill the converter with nominal liquid oxygen to the design full value as determined by weight.

b. Remove liquid oxygen in increments which will cause the indicator to read at each major dial graduation below the initial reading.

c. Record the indicator readings and weight of liquid oxygen removed for each reading.

d. The readings obtained shall not deviate respectively from the actual quantities by more than 2 percent of indication plus 4 percent of full scale indication at any of the major dial divisions.

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4.5.3 Sealing- The hermetic seal of the components shall be tested with an approved Helium Leak Detector. There shall be no leakage greater than 0.1 micron cubic foot per hour at a differential pressure of approximately one atmosphere.

4.5.4 Dialectric Test - There shall be no breakdown of insulation of any of the equipment when it is subjected to the dielectric tests specified in the applicable detail specification.

4.5.5 Power Failure - Interruption of the input power to the indicator shall cause the power failure warning device of teh indicator to indicate "OFF" until power is again applied.

4.5.6 Press to Test -A press to test circuit shall be provided in the indicator for testing the operation of the gaging system including the low level warning light by pushing a remote test button.

4.5.7 Low Liquid Level Warning Circuit - The low liquid level warning circuit shall automatically be activated at a pre-set low level value in accordance with the applicable detail specification

4.5.8 Temperature Tests -

4.5.8.1 High Temperature - The gaging components shall be subjected to a temperature of 165°F ±2° for a mimum period of four hours. At the end of this period and while wtill at this temperature, the following tests shall be conducted in accordance with the applicable detail Specification:

a. High or Low Temperature (as applicable)
b. Press to Test
c. Low Liquid Level Warning Circuit
d. Power failure

4.5.8.2 Low Temperature - The tests of 4.5.8.1 above shall be repeated except the temperature shall be -65°F ± 2°. The Gage Scale Error test shall be performed afdter the system has been energized for a minimum period of 5 minutes.

4.5.9 Vibration Test

4.5.9.1 Resonance - The gage shall be subjected to the vibration specified in paragraph 4.7.1, Procedure I, of Specification MIL-E-5272. The maximum double amplitude of the indicator pointer tip oscillation shall not exceed 1/16 inch. The test shall be conducted with the components energized and at an indicator reading of "O" and at a minimum of two additional points, as selected by the Inspector.

4.5.10 Shock Test - The gage shall be subjected to a shock test as specified in paragraph 4.15.2, Procedure II, of Specification MIL-E-5272. At the completion of this test, "Individual Tests", 4.3.1 shall be repeated. There shall be nonmechanical failure or malfunctions due to the shock test. The limiting "G" value for "Crash Safety" shall be 40 "G".

4.5.11 Acceleration Test - The gage shall be subjected to an acceleration test as specified in paragraph 4.16.2, Procedure II, of specification MIL-E-5272. The applied acceleration shall be 11 G. At the completion of this test, "Individual Tests," 4.3.1, shall be repeated. There shall be no mechanical failures or malfunctions due to the applied acceleration.

4.5.12 Radio Interference - The gage shall be subjected to Conducted and Radiated Radio interference tests in accordance with the applicable detail. specification and Specification MIL-I-6181. Test frequencies shall be in accordance with the applicable detail specification.

4.5.13 Lighting - Quality assurance provisions for lighting shall be in accordance with

4.5.14 Temperature Exposure Test

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4.5.14.1 High Temperature Exposure - The gaging system shall be subjeoted to a temperature of 165°F ± 2° for a minimum period of 48 hours. At the end of this period and while still at this temperature, the following tests shall be conducted in accordance with the applicable detail specification

a. High or Low Temperature Exposure (as applicable)
b. Press to Test
c. Low Liquid Level Warning Circuit
d. Power Failure

4.5.14.2 Low Temperature Exposure - The tests of 4.5.14.1 above shall be repeated except the temperature shall be -65°F ±2°. This test shall. be performed after the system has been energized for a minimum period of 5 minutes.

4.5.15 Humidity - The gaging system shall be subjected to the conditions of paragraph 4.4.1, Procedure I, of Specification MIL-E-5272. Following exposure, the following tests shall be conducted in accordance with the applicable detail specification:

a. Humidity
b. Press to Test
c. Low Liquid Level Warning Circuit
d. Power Failure

4.5.16 Salt Spray - The gage shall be subjected to the conditions of paragraph 4.6 of Specification MIL-E-5272.

4.6 Rejection and Retest - When tests are specified on a quantity of items that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, additional items of the lot represented shall be tested immediately to determine the cause of the failure. Individual performance tests shall not be interrupted unless the defect is of such a nature that it will seriously affect the performance or safe use of the equipment.

4.6.1 Items which have been rejected may be reworked or replaced to correct the defects and resubmitted for acceptance. Before resubmitting full particulars concerning previous rejection, and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without specific approval of the procuring agency.

50 PREPARATION FOR DELIVERY

5.1 Application - The requirements specified here apply only to direct purchases by or direct shipments to the Government.

5.2 Packaging and Packing - Items shall be packed or packaged for shipment in accordance with Group X of Specification MIL-P-6064, in a Type II, Class I container.

5.2.1 Items too large for shipment in a container, as described in Specification MIL-P-6064, shall be packaged in accordance with Specification MIL-P-5633.

5.3 Marking and labeling - Marking shall be in accordance with the requirements of Specifications MIL-P-6064, MIL-STD-129, and the applicable detail specification.

5.3.1 Precautionary Markings - The following precautionary markings shall appear on two opposite sides of each interior package, whenever practicable, depending upon the size of the carton and shall also appear on the shipping container:

"FRAGILE
DELICATE INSTRUMENTS
HANDLE WITH CARE"

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6.

6.1 Intended Use - The liquid oxygen quantity gages covered by this specification are intended to indicate the quantity of liquid oxygen contained in a liquid oxygen converter.

6.2 Definitions

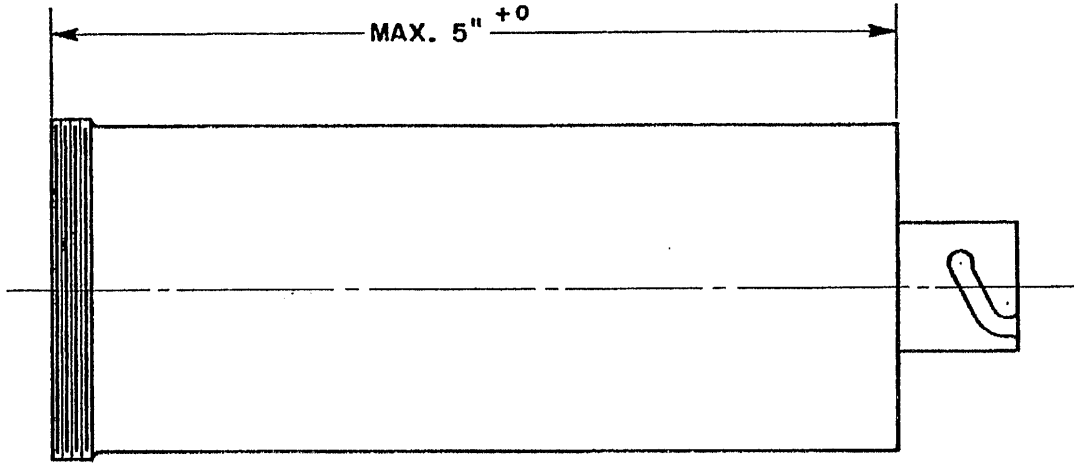
6.2.1 Procuring Agency - In this specification, the term "Procuring Agency" refers to the Bureau of Aeronautics, U. S. Navy.

6.3 Provisions for Qualification Tests - The right is reserved to reject any bids on items which have not been subjected to the required tests and found satisfactory. The attention of the manufacturer is called to this provision, and they are urged to request authorization for tests of the items which they propose to offer to the Bureau of Aeronautics under this specification. Requests for authorization of tests, together with certified test reports showing conformance with all the requirements of this specification and the manufacturer's assembly and detail drawings, and for information as to the marking and forwarding of samples should be addressed to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., the qualifying agency.

6.3.1 It is to be understood that upon receipt of the Letter of Authorization, samples shall be furnished at no cost to the Government, and that the manufacturer shall pay the transportation charges to and from the designed point where tests are to be made. In the cases of failure of the sample or samples submitted, consideration will be given to the request of the manufacturer for additional tests only after it has been clearly shown that changes have been made in the product which the Government considers sufficient to warrant additional tests.

NOTICE: When the Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation the U. S. Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in anyway supplied the said drawings, specifications, or other data is not to be inferred by triplication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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CASE IN ACCORDANCE WITH AND10412

FIGURE 1.
INDICATOR GAGE,
LIQUID OXYGEN QUANTITY, CAPACITANCE.

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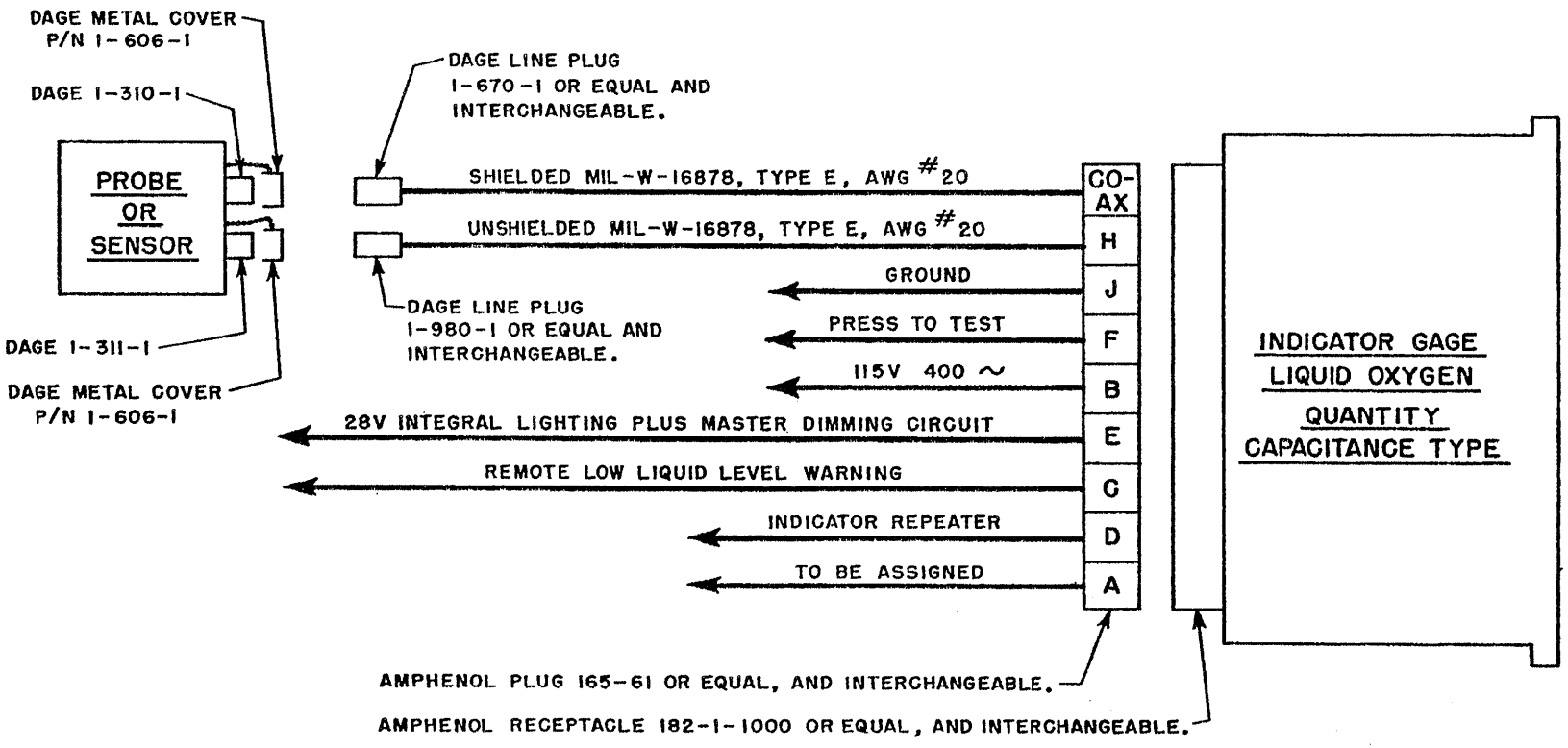


FIGURE 2
GAGING SYSTEM ARRANGEMENT.

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